

→ In java script we have to type of code

- ① Synchronous ← Pure js code
- ② Asynchronous

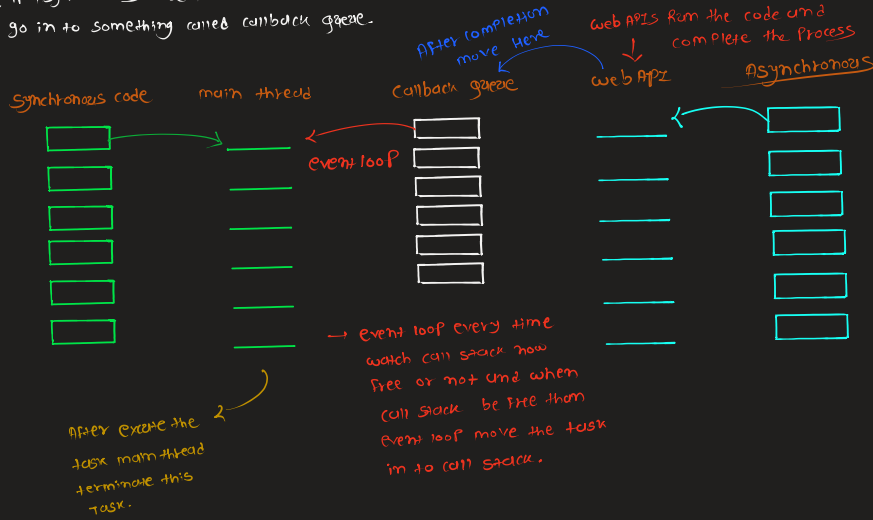
↳ Promises
callbacks
setTimeout
setInterval
Sync await

Now → js is single threaded.
js at a time only one operation should perform.
Ex: octa core processor
↳ js at a time 1 core use.

→ At that time Synchronous code is run if Asynchronous code come before that time then Asynchronous code go in to web API.

- ① main thread :- Firstly Synchron code run mean which code come in to main stack that code run.
- ② web API :- Browser provided many features and this features we can use in java script language. But in Actual these feature not provides provided java script language.

→ Synchronous code will run firstly and at that time if Asynchronous code present then this code go in to something called callback queue.



Ex: `setTimeout(function() { log("Hello"); }, 3000);`

→ So, this code go in to web API run code and wait for 3000ms because setTimeout is 3000ms and after that this task go in to callback queue.

→ one more important think before task go in to main thread task go in to call stack.

